

$$a_n = 3a_{n-1} - 2n + 1, a_1 = 2$$

$$a_n = pa_{n-1} + q$$

$$\frac{a_n}{3^n} = \frac{a_{n-1}}{3^{n-1}} - \frac{2n-1}{3^n}$$

$$b_n = b_{n-1} - \frac{2n-1}{3^n}$$

$$b_n - b_1 = -\sum_{k=2}^n \frac{2k-1}{3^k}$$

$$\left\{ \begin{aligned} S &= \sum_{k=2}^n \frac{2k-1}{3^k} \\ \frac{S}{3} &= \sum_{k=2}^n \frac{2k-1}{3^{k+1}} = \sum_{j=3}^{n+1} \frac{2j-3}{3^j} \end{aligned} \right.$$

$$\left\{ \begin{aligned} S &= \sum_{k=2}^n \frac{1}{3^{k-1}} + \frac{1}{6} - \frac{2n-1}{2 \cdot 3^n} \\ \frac{2}{3} S &= \sum_{k=2}^n \frac{2k-1}{3^k} - \left(\sum_{k=2}^n \frac{2k-3}{3^k} - \frac{1}{9} + \frac{2n-1}{3^{n+1}} \right) \end{aligned} \right. = \sum_{k=2}^n \frac{2}{3^k} + \frac{1}{9} - \frac{2n-1}{3^{n+1}}$$

$$\frac{a_n}{3^n} = \frac{2}{3} - \sum_{k=2}^n \frac{1}{3^{k-1}} - \frac{1}{6} + \frac{2n-1}{2 \cdot 3^n} = \frac{2}{3} - \frac{1}{3} \left(\frac{1 - (\frac{1}{3})^{n-1}}{1 - \frac{1}{3}} \right) - \frac{1}{6} + \frac{2n-1}{2 \cdot 3^n}$$

$$= \cancel{\frac{2}{3}} - \cancel{\frac{1}{3}} + \frac{1}{2 \cdot 3^{n-1}} - \cancel{\frac{1}{6}} + \frac{n}{3^n} - \frac{1}{2 \cdot 3^n} = \frac{n}{3^n} + \frac{1}{3^n}$$

$$\boxed{a_n = n+1}$$

$$n+1 = 3(n) - 2n + 1 = n+1$$

$$a_1 = 2$$

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